Domain & Bidding Type Guide

Forward vs reverse bidding, compared for the three domains, with a suggestion for the team

1. Two choices to make

Domain = what is auctioned and who uses it (General, Campus, or Institutional Surplus).
Bidding type = who competes. In **forward** bidding, buyers compete and the price goes **up**. In **reverse** bidding, sellers compete and the price goes **down**.

2. Campus domain: forward vs reverse in simple steps

	Forward bidding (campus)	Reverse bidding (campus)
Who starts	A seller lists an item, e.g. a calculator with start price Rs. 500.	A buyer posts a need, e.g. "Need a calculator, max budget Rs. 600, by Friday."
Who competes	Buyers raise their bids. Each bid must be higher than the current one.	Sellers make offers. Each new offer must be lower than the current lowest.
How it ends	At the deadline the highest valid bid wins automatically.	At the deadline the buyer picks the best offer (not always the lowest: condition and pickup also matter).
Saved in database	Result and transaction status.	Chosen offer and transaction status.
Fake activity rules	- College email login only - Bids cannot be withdrawn - Seller cannot bid on own item - Minimum raise per bid - Bid history visible, no editing	- College email login only - Offers cannot be withdrawn once chosen - Seller can only lower own offer - Buyer confirms receipt; sellers get ratings - Ratings shown next to offers
Main risk	Price gets pushed up, which favours the seller. Fake bids can inflate it.	Too few sellers for niche requests. A seller may offer too low and not deliver.
How to reduce risk	Show full history, limit bids on own items, keep rules visible.	Keep requests open 2-3 days. Let seniors and clubs offer too. Use several sellers in demo data.

3. Pros and cons in each domain

Domain	Forward bidding	Reverse bidding
A. General Platform	**Pros** + Natural fit, many buyers + Easiest to explain and demo + Simple bid check **Cons** - Very common idea - Fake bids and inflated prices - Favours the seller	**Pros** + Unique and buyer-friendly **Cons** - Requests for "any product" are vague and hard to compare - Needs many sellers across many categories - Unfamiliar to most users
B. Campus Exchange	**Pros** + Easy to build, cheap items with many interested buyers + Very relatable for students + Simple demo data **Cons** - Favours sellers, not buyers - Least unique - Fake bids still need rules	**Pros** + Buyer-focused and unique + Students post real needs + Lower prices for students + Buyer chooses winner, so logic stays simple **Cons** - Few sellers per request - Seller trust and delivery - Needs sample sellers for a good demo
C. Institutional Surplus	**Pros** + Best fit: retired equipment goes to the highest bidder + Approved bidders reduce fake bids + Strong database design **Cons** - More rules: approvals, lots, admin - Must simulate the institution	**Pros** + Works for buying supplies, but that is a different problem **Cons** - Does not match surplus selling - Quote comparison and approvals are heavy - Hardest to finish in one month

Overall: forward vs reverse

	Forward	Reverse
Strengths	Popular, easy to explain, safest to finish	Buyer-focused, unique, stands out in the demo
Weaknesses	Common idea, favours sellers, fake bids	Needs enough sellers, more trust rules
Effort	Lower	Slightly higher, but manageable if kept simple

4. Our suggestion

Domain: Campus Auction & Resource Exchange. It is a specific problem, easy to explain, easy to create demo data for, and the scope is manageable. General is too common, and Surplus has too many approval rules for one month.

Bidding type: reverse only, in a simple version (buyer posts a request, sellers offer lower prices, buyer chooses). This matches the goal of favouring buyers and makes the project different. Do not build both models.

Safety net: forward bidding is the lower-risk option. The tables are nearly the same (requests = auctions, offers = bids). If the core flow is not working by the end of week 2, switch to forward.

5. Questions for the first meeting

- Can we explain the problem in one line?
- Can we model it in MySQL with about 6 tables?
- Can we finish the core flow in under one month?
- Do we have enough sample sellers (reverse) or buyers (forward) to make the demo look real?